

Supplementary Material for “SlotRAG: Budget-Feasible Physical Planning over Typed Evidence Plans for Multi-Hop Retrieval-Augmented Generation”

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1 Pre-registration History (H-STRUCT)

The evaluation follows a frozen pre-registered protocol. The hypothesis chain is archived as: H-STRUCT-1 (budget-feasibility diagnosis; census of deep eligible plans), H-STRUCT-2 (paired matched-budget confirmation of the flat planner on deep plans), H-STRUCT-3 (structure-gate dispatch necessity, exploratory replay), and H-STRUCT-4 (independent shallow-regime confirmation). Pre-registration documents, amendments, and final reports are versioned in the repository under `research/H-STRUCT-1`. Table 1 summarizes each hypothesis, its evidence label, and outcome.

Hyp.	Question	Evidence level	Outcome
H-STRUCT-1	Feasibility diagnosis recall/precision	Confirmatory (n=350)	Confirmed
H-STRUCT-2	Flat vs. static on deep plans	Confirmatory (n=350)	Confirmed
H-STRUCT-3	Gate dispatch necessity	Exploratory (n=8,632)	Supported (exploratory)
H-STRUCT-4	Independent shallow confirmation	Pre-registered	Infeasible (payload gap)

Table 1: H-STRUCT hypothesis history.

2 Statistical Protocol

All paired comparisons share the same compiled plans, questions, and seeds. The primary test is the two-sided exact McNemar test (binomial exact test on discordant pairs). Confidence intervals are paired bootstrap percentiles (10,000 iterations, seed 2027, full- N question-level resampling). Dataset heterogeneity is assessed with Holm-corrected tests across datasets and strata. Budget-exceeded items score 0.0 EM because evaluation uses exact-match scoring and budget-exhausted executions without a final answer receive EM = 0. The confirmatory budget is $B = 8$ (`max_steps = 8`, `max_retrieval_calls = 8`, `max_llm_calls = 96`); an earlier permissive exploratory trace used higher limits and is explicitly labeled *exploratory*.

3 H-STRUCT-4: Infeasible Confirmation Attempt

A confirmatory shallow-regime matched-budget evaluation was pre-registered with $n = 1,428$ required (power analysis, $\alpha = 0.05$, 80% power). The V1.2 frozen census did not persist full SlotPlan JSON for the 6,133 shallow questions—only structural metadata and plan hashes—and the protocol prohibited re-compilation. Zero answers were generated; the attempt produced no counterexamples. The gate’s shallow-regime benefit therefore remains supported only by exploratory evidence; an independent confirmation under the frozen matched-budget protocol remains an open requirement.

4 Extended Dataset-Stratum Tables

Stratum	Flat – Static	95% CI	Regime
Deep (hops ≥ 2 , n=350)	+0.0771	[+0.049, +0.109]	Confirmatory
Shallow (hops < 2 , n=8,085)	−0.0210	[−0.027, −0.015]	Exploratory
Shallow 2Wiki hops-0	−0.045	–	Exploratory (secondary)
HotpotQA shallow	≈ 0	–	Exploratory
MuSiQue shallow	≈ 0	–	Exploratory

Table 2: Stratum-level flat-vs-static EM effects.

5 Additional Failure Analysis

Static budget-exhausted items had average EM below 0.01, confirming that partial evidence rarely yields correct answers on deep multi-hop questions. Budget exhaustion is structural: it arises from the mismatch between local slot allocations and the global budget (41.7% of deep static executions), not from retrieval failures or generation errors. The flat planner eliminates these events by construction ($\sum_s a_s \leq B$).

6 Reproducibility and Provenance

Frozen artifacts (sample registries, exposure audits, plan hashes, and experiment ledgers) are archived under `research/` and `runs/`. The reproduction guide in `README.md` distinguishes fully reproducible analyses (statistics, offline replay, figures, confusion matrices) from pipeline steps that require external model/API access.